

CP2.4 — Volumes of Revolution (Further)

1. Review — Volume of Revolution (Cartesian)

Rotating $y = f(x)$ about the **x-axis** between $x = a$ and $x = b$:

$$V = \pi \int_a^b y^2 dx$$

Rotating $x = g(y)$ about the **y-axis** between $y = c$ and $y = d$:

$$V = \pi \int_c^d x^2 dy$$

2. Volume of Revolution — Parametric Equations

For a curve given parametrically as $x = f(t)$, $y = g(t)$, rotated about the **x-axis**:

$$V = \pi \int_{t_1}^{t_2} y^2 \frac{dx}{dt} dt$$

where t_1 and t_2 are the parameter values corresponding to $x = a$ and $x = b$.

Exam tip: The formula uses $\frac{dx}{dt} dt$ as the element of integration (the substitution $dx = \frac{dx}{dt} dt$). Make sure $\frac{dx}{dt} > 0$ over the range — if not, you may need to take the magnitude or swap limits.

Example: Find the volume generated when the arc of the curve $x = t^2$, $y = 2t$, $0 \leq t \leq 2$, is rotated through 2π radians about the x-axis.

$$\frac{dx}{dt} = 2t, y^2 = 4t^2.$$

$$V = \pi \int_0^2 4t^2 \cdot 2t dt = \pi \int_0^2 8t^3 dt = \pi [2t^4]_0^2 = \pi \cdot 32 = 32\pi$$

Example: The curve is defined by $x = \cos^3 \theta$, $y = \sin^3 \theta$ for $0 \leq \theta \leq \pi/2$ (an astroid). Find the volume of revolution about the x-axis.

When $\theta = 0$: $x = 1$. When $\theta = \pi/2$: $x = 0$. Since x decreases as θ increases, $\frac{dx}{d\theta} < 0$; swap limits:

$$\frac{dx}{d\theta} = -3 \cos^2 \theta \sin \theta, y^2 = \sin^6 \theta$$

$$V = \pi \int_0^{\pi/2} \sin^6 \theta \cdot 3 \cos^2 \theta \sin \theta d\theta = 3\pi \int_0^{\pi/2} \sin^7 \theta \cos^2 \theta d\theta$$

Using the reduction formula or substitution $u = \cos \theta$:

$$= 3\pi \int_0^1 (1 - u^2)^3 u^2 (-du) \cdot (-1)$$

(With $u = \cos \theta$, $du = -\sin \theta d\theta$, limits $\theta = 0 \rightarrow u = 1$, $\theta = \pi/2 \rightarrow u = 0$):

$$= 3\pi \int_0^1 (1 - u^2)^3 u^2 du$$

Expanding $(1 - u^2)^3 = 1 - 3u^2 + 3u^4 - u^6$:

$$= 3\pi \int_0^1 (u^2 - 3u^4 + 3u^6 - u^8) du = 3\pi \left[\frac{1}{3} - \frac{3}{5} + \frac{3}{7} - \frac{1}{9} \right] = 3\pi \cdot \frac{105 - 189 + 135 - 35}{315} = 3\pi \cdot \frac{16}{315} =$$

3. Surface Area of Revolution

Rotating $y = f(x)$ about the **x-axis** from a to b :

$$S = 2\pi \int_a^b y ds, \quad \text{where } ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

So:

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

3.1 Parametric Form for Surface Area

For $x = f(t)$, $y = g(t)$, rotating about the x-axis:

$$S = 2\pi \int_{t_1}^{t_2} y \, ds, \quad ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$S = 2\pi \int_{t_1}^{t_2} y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example: Find the surface area generated when $x = t^2$, $y = 2t$, $0 \leq t \leq 3$ is rotated about the x-axis.

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 2.$$

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{4t^2 + 4} = 2\sqrt{t^2 + 1}$$

$$S = 2\pi \int_0^3 2t \cdot 2\sqrt{t^2 + 1} \, dt = 8\pi \int_0^3 t\sqrt{t^2 + 1} \, dt$$

Let $u = t^2 + 1$, $du = 2t \, dt$:

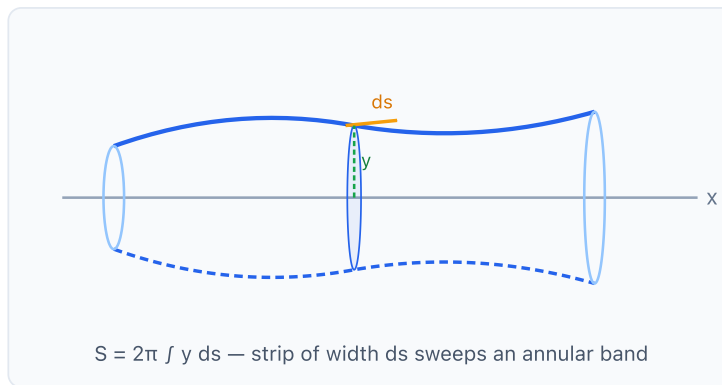
$$= 8\pi \int_1^{10} \frac{\sqrt{u}}{2} du = 4\pi \left[\frac{2}{3} u^{3/2} \right]_1^{10} = \frac{8\pi}{3} (10^{3/2} - 1) = \frac{8\pi(10\sqrt{10} - 1)}{3}$$

3.2 Rotation about the y-axis

Rotating about the **y-axis**: replace y with x in the formula:

$$S = 2\pi \int x \, ds$$

3.3 Diagram — Surface of Revolution



4. Arc Length

Cartesian: $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

Parametric: $L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

Example: Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

$$\frac{dy}{dx} = x^{1/2}, \text{ so } \sqrt{1 + (dy/dx)^2} = \sqrt{1 + x}.$$

$$L = \int_0^3 \sqrt{1+x} \, dx = \left[\frac{2}{3}(1+x)^{3/2} \right]_0^3 = \frac{2}{3}(8-1) = \frac{14}{3}$$

Example: Find the arc length of $x = t^2/2$, $y = t^3/3$ from $t = 0$ to $t = 1$.

$$\frac{dx}{dt} = t, \frac{dy}{dt} = t^2.$$

$$L = \int_0^1 \sqrt{t^2 + t^4} \, dt = \int_0^1 t\sqrt{1+t^2} \, dt = \left[\frac{1}{3}(1+t^2)^{3/2} \right]_0^1 = \frac{1}{3}(2\sqrt{2}-1)$$

5. Volume Between Curves (Washer Method)

When rotating the region between $y = f(x)$ and $y = g(x)$ (with $f > g \geq 0$) about the x-axis:

$$V = \pi \int_a^b [f(x)^2 - g(x)^2] dx$$

Example: Find the volume when the region between $y = x + 1$ and $y = x^2 - 1$ (for $x \in [0, 2]$) is rotated about the x-axis.

Here $f(x) = x + 1$, $g(x) = x^2 - 1$ (check: at $x = 0$, $f = 1 > g = -1$).

Note: $g(x) = x^2 - 1 < 0$ for $0 \leq x < 1$, so the region crosses the x-axis there. For $0 \leq x \leq 1$, $|g(x)| \leq f(x)$, so each slice is a solid disc of radius f (no hole); the washer formula applies only for $1 \leq x \leq 2$.

$$\begin{aligned} V &= \pi \int_0^1 (x+1)^2 dx + \pi \int_1^2 [(x+1)^2 - (x^2-1)^2] dx \\ &= \pi \left[\frac{(x+1)^3}{3} \right]_0^1 + \pi \int_1^2 (-x^4 + 3x^2 + 2x) dx \\ &= \frac{7\pi}{3} + \pi \left[-\frac{x^5}{5} + x^3 + x^2 \right]_1^2 = \frac{7\pi}{3} + \frac{19\pi}{5} = \frac{92\pi}{15} \end{aligned}$$

6. Summary of Formulas

Quantity	Cartesian	Parametric
Volume (x-axis)	$\pi \int y^2 dx$	$\pi \int y^2 \frac{dx}{dt} dt$
Volume (y-axis)	$\pi \int x^2 dy$	$\pi \int x^2 \frac{dy}{dt} dt$
Arc length	$\int \sqrt{1 + (y')^2} dx$	$\int \sqrt{\dot{x}^2 + \dot{y}^2} dt$
Surface area (x-axis)	$2\pi \int y ds$	$2\pi \int y \sqrt{\dot{x}^2 + \dot{y}^2} dt$
Surface area (y-axis)	$2\pi \int x ds$	$2\pi \int x \sqrt{\dot{x}^2 + \dot{y}^2} dt$

Exam tip: The Edexcel formula booklet gives the volume formula but NOT the surface area formula — you must know $S = 2\pi \int y ds$ and the expression for ds . For parametric curves, always compute

$\sqrt{(\dot{x})^2 + (\dot{y})^2}$ first and simplify before integrating.

7. Exam-Style Practice Questions

1. **[Parametric volume]** The curve C is defined parametrically by $x = 3t^2$, $y = 6t$ for $t \geq 0$. The region bounded by C and the x-axis for $0 \leq x \leq 12$ is rotated 2π about the x-axis. Find the volume.
2. **[Surface area]** The curve $y = \frac{1}{4}x^2 - \frac{1}{2}\ln x$ is rotated about the y-axis from $x = 1$ to $x = 2$. Find the exact surface area.
3. **[Arc length]** Find the arc length of $x = e^t \cos t$, $y = e^t \sin t$ from $t = 0$ to $t = \pi$.
4. **[Washer]** The region enclosed by $y = \sin x$ and $y = \cos x$ for $\pi/4 \leq x \leq 5\pi/4$ is rotated about the x-axis. Set up (but do not evaluate) the integral for the volume.
5. **[Parametric surface area]** Find the surface area when the circle $x = a \cos t$, $y = a \sin t$ ($0 \leq t \leq \pi$) is rotated about the x-axis. (Answer: $4\pi a^2$ — the sphere.)